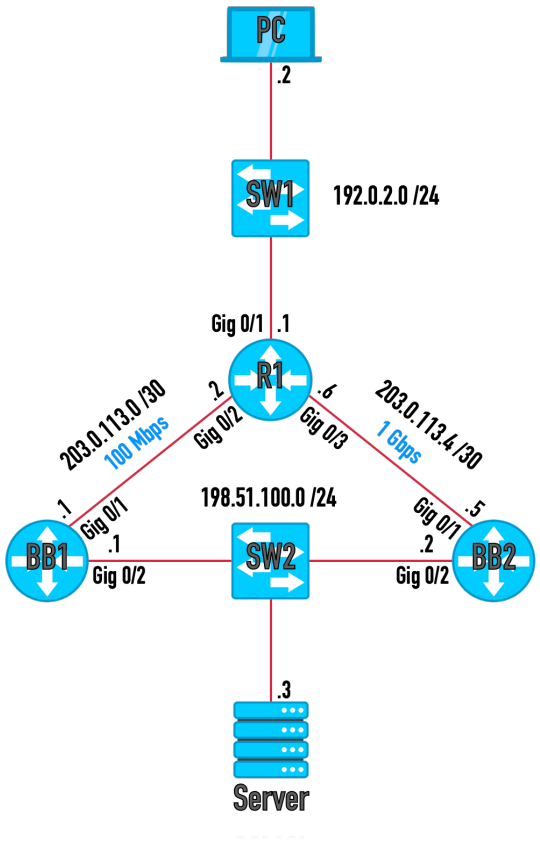
IP SLA Path Selection

Topology



Initial Configuration Commands

PC:

```
enable
conf t
host PC
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
line con 0
logging synchronous
```

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```
exit
no ip domain-lookup
no ip routing
int gig 0/1
ip address 192.0.2.2 255.255.255.0
no shut
ip default-gateway 192.0.2.1
end
copy run star
```

R1:

```
enable
conf t
host R1
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
line con 0
logging synchronous
exit
no ip domain-lookup
int gig 0/1
ip address 192.0.2.1 255.255.255.0
no shut
int gig 0/2
ip address 203.0.113.2 255.255.255.252
ip ospf cost 100
no shut
int gig 0/3
ip address 203.0.113.6 255.255.255.252
no shut
router ospf 1
network 0.0.0.0 0.0.0.0 area 0
auto-cost reference-bandwidth 10000
end
copy run star
```

BB1:

```
enable
conf t
```

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```
host BB1
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
exit
no ip domain-lookup
lldp run
int gig 0/1
ip address 203.0.113.1 255.255.255.252
ip ospf cost 100
no shut
int gig 0/2
ip address 198.51.100.1 255.255.255.0
speed 100
no shut
router ospf 1
network 0.0.0.0 0.0.0.0 area 0
auto-cost reference-bandwidth 10000
end
copy run star
```

BB2:

```
enable
conf t
host BB2
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
exit
no ip domain-lookup
lldp run
```

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```
int gig 0/1
ip address 203.0.113.5 255.255.255.252
speed 100
no shut
int gig 0/2
ip address 198.51.100.2 255.255.255.0
no shut
router ospf 1
network 0.0.0.0 0.0.0.0 area 0
auto-cost reference-bandwidth 10000
end
copy run star
```

Server:

```
enable
conf t
host SERVER
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
exit
no ip domain-lookup
lldp run
int gig 0/1
ip address 198.51.100.3 255.255.255.0
no shut
router ospf 1
network 0.0.0.0 0.0.0.0 area 0
auto-cost reference-bandwidth 10000
end
copy run star
```

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Lab Tasks

1. Let’s do a traceroute from the PC to the SERVER to see what path the PC takes to reach the SERVER.
2. On R1, let's configure an IP SLA Operation to switch to BB1 if the delay becomes too high. Let’s set our locally significant number to “1”.
3. Next, we are going to use an ICMP Echo message sent from the Gig 0/1 interface on R1 to the Gig 0/1 interface on BB2.
4. Let's set the probe frequency to 5 seconds for testing network delay.
5. Let’s configure the threshold to specify that if the delay exceeds this value, the backup route should be used. Set the threshold to 100 ms.
6. Let’s set up our operation on R1 to start “now”, and let’s have it run “forever.”
7. Let’s associate this IP SLA 1 Operation with a tracking object. In the tracking object, we can specify how quickly we want to switch states if the delay exceeds or falls below the threshold. We'll set the delay to be 10 seconds for both down and up.
8. Let's set up a static route so that traffic from R1 to the 198.51.100.0/24 network uses the next-hop IP address 203.0.113.5 (which is BB2). However, we only want to use this route if our tracking object is UP—which means, the round-trip delay from R1 to BB2 must be less than 100 ms.
9. Now, let’s create a backup route that gets us to the 198.51.100.0 /24 network via BB1. We need to make this static route less preferable by setting the administrative distance to 2.
10. On R1, let’s see what route we are using to reach the 198.51.100.0 /24 network by using the “show ip route” command.
11. Let’s verify that the tracking object is up by using the “show track 1” command. Let’s also verify our SLA Operation by using the “show ip sla statistics” command.
12. Unfortunately, once we create an IP SLA operation, it cannot be edited. To adjust the threshold to 1 millisecond and trigger a down state, we’ll need to delete the existing IP SLA operation and then recreate it with the new threshold of 1 millisecond.
13. Now, if we check our tracking state, we will probably see that it is down because our delay is over our configured threshold.
14. Let’s verify that we are now going to go through BB1 to reach the 198.51.100.0 /24 network.
15. Finally, on the PC, let’s do a traceroute command to see what path we will take to reach the server.

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Solution

Step 1: Let's do a traceroute from the PC to the SERVER to see what path the PC takes to reach the SERVER.

```
PC>en
PC#traceroute 198.51.100.3
Type escape sequence to abort.
Tracing the route to 198.51.100.3
VRF info: (vrf in name/id, vrf out name/id)
  1 192.0.2.1 1014 msec 11 msec 12 msec
  2 203.0.113.5 17 msec 15 msec 18 msec
  3 198.51.100.3 29 msec *  16 msec
```

(##Since this network topology is configured with OSPF, the PC will select the path with the lowest cost, which happens to be BB2 because it is a Gig link.)

Step 2: On R1, let's configure an IP SLA Operation to switch to BB1 if the delay becomes too high. Let's set our locally significant number to "1".

```
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip sla 1
```

Step 3: Next, we are going to use an ICMP Echo message sent from the Gig 0/1 interface on R1 to the Gig 0/1 interface on BB2.

```
R1(config-ip-sla)#?
IP SLAs entry configuration commands:
  dhcp          DHCP Operation
  dns           DNS Query Operation
  ethernet      Ethernet Operations
  exit          Exit Operation Configuration
  ftp           FTP Operation
  http          HTTP Operation
  icmp-echo     ICMP Echo Operation
  icmp-jitter   ICMP Jitter Operation
  mpls          MPLS Operation
  path-echo     Path Discovered ICMP Echo Operation
```

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```
  path-jitter  Path Discovered ICMP Jitter Operation
  tcp-connect  TCP Connect Operation
  udp-echo     UDP Echo Operation
  udp-jitter   UDP Jitter Operation
  voip         Voice Over IP Operation
```

```
R1(config-ip-sla)#icmp-echo 203.0.113.5 source-ip 192.0.2.1
```

(##Icmp-echo is another way of saying that we are going to be sending a ping packet.)

Step 4: Let's set the probe frequency to 5 seconds for testing network delay.

```
R1(config-ip-sla-echo)#frequency 5
```

Step 5: Let's configure the threshold to specify that if the delay exceeds this value, the backup route should be used. Set the threshold to 100 ms.

```
R1(config-ip-sla-echo)#threshold 100
R1(config-ip-sla-echo)#exit
```

Step 6: Let's set up our operation on R1 to start "now", and let's have it run "forever."

```
R1(config)#ip sla schedule 1 life forever start-time now
```

Step 7: Let's associate this IP SLA 1 Operation with a tracking object. In the tracking object, we can specify how quickly we want to switch states if the delay exceeds or falls below the threshold. We'll set the delay to be 10 seconds for both down and up.

```
R1(config)#track 1 ip sla 1
R1(config-track)#delay down 10 up 10
R1(config-track)#exit
```

(##This means, "Don't transition to a down state unless the condition of exceeding the threshold has persisted for 10 seconds or more. Conversely, if the value briefly spikes from 98 to 105 for just a second, it isn't a sufficient reason to switch to an up state".)

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Step 8: Let's set up a static route so that traffic from R1 to the 198.51.100.0/24 network uses the next-hop IP address 203.0.113.5 (which is BB2). However, we only want to use this route if our tracking object is UP—which means, the round-trip delay from R1 to BB2 must be less than 100 ms.

```
R1(config)#ip route 198.51.100.0 255.255.255.0 203.0.113.5 track 1
```

(##We are saying that under normal conditions (and tracking object number 1 is up), we want to use BB2 as our next hop router.)

Step 9: Now, let's create a backup route that gets us to the 198.51.100.0 /24 network via BB1. We need to make this static route less preferable by setting the administrative distance to 2.

```
R1(config)#ip route 198.51.100.0 255.255.255.0 203.0.113.1 2
R1(config)#end
```

(##Remember, by default, the administrative distance of a static route is 1.)

Step 10: On R1, let's see what route we are using to reach the 198.51.100.0 /24 network by using the "show ip route" command.

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static
route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      192.0.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.0.2.0/24 is directly connected, GigabitEthernet0/1
L        192.0.2.1/32 is directly connected, GigabitEthernet0/1
S     198.51.100.0/24 [1/0] via 203.0.113.5
      203.0.113.0/24 is variably subnetted, 4 subnets, 2 masks
C        203.0.113.0/30 is directly connected, GigabitEthernet0/2
L        203.0.113.2/32 is directly connected, GigabitEthernet0/2
C        203.0.113.4/30 is directly connected, GigabitEthernet0/3
L        203.0.113.6/32 is directly connected, GigabitEthernet0/3
```

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(##You can see here that we are using BB2 to reach the 198.51.100.0 /24 network.)

Step 11: Let's verify that the tracking object is up by using the "show track 1" command. Let's also verify our SLA Operation by using the "show ip sla statistics" command.

```
R1#show track 1
Track 1
  IP SLA 1 state
  State is Up
    1 change, last change 00:11:18
  Delay up 10 secs, down 10 secs
  Latest operation return code: OK
  Latest RTT (millisecs) 5
  Tracked by:
    Static IP Routing 0
R1#show ip sla statistics
IPSLAs Latest Operation Statistics

IPSLA operation id: 1
        Latest RTT: 3 milliseconds
Latest operation start time: 20:50:25 UTC Sat Aug 10 2024
Latest operation return code: OK
Number of successes: 151
Number of failures: 0
Operation time to live: Forever
```

(##Here you can see that our tracking object is up.)
(##Our latest round trip delay in this demonstration is 5 milliseconds. Based on this information that we pulled from this demonstration, we can see that we have not gone over 100 milliseconds, meaning that our IP SLA Operation hasn't routed traffic over to BB1.)
(##Here, we can observe where we've configured our SLA operation to run forever.)

Step 12: Unfortunately, once we create an IP SLA operation, it cannot be edited. To adjust the threshold to 1 millisecond and trigger a down state, we'll need to delete the existing IP SLA operation and then recreate it with the new threshold of 1 millisecond.

```
R1#conf t
```

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```
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#no ip sla 1
R1(config)#ip sla 1
R1(config-ip-sla)#icmp-echo 203.0.113.5 source-ip 192.0.2.1
R1(config-ip-sla-echo)#frequency 5
R1(config-ip-sla-echo)#threshold 1
R1(config-ip-sla-echo)#exit
R1(config)#ip sla schedule 1 life forever start-time now
R1(config)#end
```

(##Here is where we set our threshold to 1 millisecond.)

Step 13: Now, if we check our tracking state, we will probably see that it is down because our delay is over our configured threshold.

```
R1#show track 1
Track 1
  IP SLA 1 state
  State is Down
    2 changes, last change 00:01:46
  Delay up 10 secs, down 10 secs
  Latest operation return code: Over threshold
  Latest RTT (millisecs) 4
  Tracked by:
    Static IP Routing 0
```

(##Notice that the state of our tracking object is now down because we have surpassed our threshold of 1 millisecond.)

Step 14: Let's verify that we are now going to go through BB1 to reach the 198.51.100.0 /24 network.

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static
route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
```

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```
Gateway of last resort is not set

      192.0.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.0.2.0/24 is directly connected, GigabitEthernet0/1
L        192.0.2.1/32 is directly connected, GigabitEthernet0/1
S     198.51.100.0/24 [2/0] via 203.0.113.1
      203.0.113.0/24 is variably subnetted, 4 subnets, 2 masks
C        203.0.113.0/30 is directly connected, GigabitEthernet0/2
L        203.0.113.2/32 is directly connected, GigabitEthernet0/2
C        203.0.113.4/30 is directly connected, GigabitEthernet0/3
L        203.0.113.6/32 is directly connected, GigabitEthernet0/3
```

Step 15: Finally, on the PC, let's do a traceroute command to see what path we will take to reach the server.

```
PC#traceroute 198.51.100.3
Type escape sequence to abort.
Tracing the route to 198.51.100.3
VRF info: (vrf in name/id, vrf out name/id)
  1 192.0.2.1 19 msec 18 msec 21 msec
  2 203.0.113.1 17 msec 15 msec 15 msec
  3 198.51.100.3 24 msec *  13 msec
```

(##Notice that we are now going through BB1 to reach the server.)

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